

Customer Churn Prediction & Risk Segmentation Report

1. Introduction

Customer churn significantly impacts revenue in subscription-based businesses. This project focuses on predicting customer churn using machine learning and segmenting customers based on risk levels to enable proactive retention strategies.

2. Objective

- Predict customers likely to churn
 - Identify key churn drivers
 - Compare machine learning models
 - Segment customers into risk categories
 - Provide actionable business insights
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3. Methodology

The dataset was analyzed for missing values, data types, and patterns. Categorical variables were encoded using One-Hot Encoding, and numerical features were scaled. New features such as *ChargesPerMonth* and *SeniorWithNoSupport* were created to improve model performance.

Three models were trained:

- Logistic Regression
- Random Forest
- Gradient Boosting

Models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

4. Model Performance

The **Gradient Boosting model** performed best with the highest ROC-AUC score and a good balance between precision and recall, making it the most reliable for churn prediction.

5. Key Findings

- Month-to-month contracts show the highest churn
 - Customers with low tenure are more likely to leave
 - Higher monthly charges increase churn probability
 - High-risk customers typically have short tenure and no long-term contracts
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6. Customer Segmentation

Customers were grouped based on churn probability:

-  High Risk ($\geq 70\%$)
 -  Medium Risk (40–69%)
 -  Low Risk ($< 40\%$)
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7. Recommendations

- Encourage long-term contracts through incentives
 - Improve onboarding for new customers
 - Target high-risk customers with personalized offers
 - Review pricing strategies for high-charge customers
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8. Limitations

- No behavioral or usage data included
 - Model based only on historical data
 - Performance can improve with more features and tuning
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9. Conclusion

The project successfully predicts customer churn and segments customers by risk. These insights can help businesses reduce churn and improve customer retention strategies.